German Power Fair Value — Day-Ahead Forecasting Pipeline

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Market: DE-LU Day-Ahead | **Submission:** July 2026

The German DA market in 2024 is defined by two opposing forces: renewable cannibalization driving 457 negative-price hours (5.2% of the year) as solar penetration hit 17% of generation, and Dunkelflaute events in November and December 2024 that drove German power prices to intraday peaks of €820/MWh and €900/MWh, with day-ahead prices reaching €145/MWh and €175/MWh respectively — the highest in 18 years. Any credible fair-value system must model both regimes. This pipeline does.

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1. Data Ingestion and Quality

Source: SMARD (smard.de) — Bundesnetzagentur electricity market platform. Data retrieved from ENTSO-E under EU Regulation 543/2013. Equivalent data quality, no API key required, fully public.

Programmatic API endpoints:

Series	Filter ID	Endpoint Pattern	Unit
DA Price DE-LU	4169	/chart_data/4169/DE-LU/	EUR/MWh
Total Load	410	/chart_data/410/DE/	MW
Wind Onshore	123	/chart_data/123/DE/	MW
Wind Offshore	3791	/chart_data/3791/DE/	MW
Solar PV	125	/chart_data/125/DE/	MW

Base URL: `https://www.smard.de/app/chart_data/{filter}/{region}/{filter}_{region}_hour_{ts}.json`
Response: `[[unix_ms, value], ...]` arrays converted to UTC-indexed hourly Series.

Dataset: 26,304 hours, 2022-01-01 to 2024-12-31.
Timezone: UTC internal storage; CET/CEST for display. Spring-forward gap interpolated; autumn duplicate averaged. Assert: exactly 8,784 rows for 2024 — passes.

Structural breaks: post_ukraine_war (2022-02-24); post_nuclear_phaseout (2023-04-15, DE nuclear = 0 MW).

LLM Data QA (Gemini 2.0 Flash): 20 electricity-market-specific validation rules proposed programmatically and executed as Python boolean conditions (solar zero 20:00–05:00 CET, load 20–100 GW, price −500 to 3,000 EUR/MWh, nuclear = 0 after phase-out, Dunkelflaute logic). QA score: 97.1/100 — PASS. This reduces manual rule authoring from hours to seconds while producing domain-specific rules a generalist engineer would miss.

2024 verified facts: Mean DA price €78.51/MWh; negative hours 457 (5.2%); Dunkelflaute detected: Nov 2–7 2024 (111 Dunkelflaute hours within 144h window), Dec 12–14 2024 (45

Dunkelflaute hours within 72h window).

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2. Forecasting and Model Validation

Target: Hourly DA prices for delivery day + 1. Delivery-period aggregates derived from the hourly distribution.

Fundamental insight: German prices are driven by residual load = load − wind − solar. High residual load → gas marginal → prices spike. Low or negative → renewables marginal → prices collapse. Features capture this at multiple timescales.

67 features across 8 groups: fundamental supply/demand (11, led by residual_load), fuel/carbon (7, including clean spark spread), calendar with cyclic encoding (16), lag features shifted minimum 24h (8, price_lag_168h is strongest predictor at $r = 0.71$), rolling statistics (10), interaction terms (6), regime features (8, including dunkelflaute_severity), and structural break dummies (3). Leakage assertion: no feature shows correlation above 0.95 with future target.

Results:

Model	2024 MAE	Skill vs Naive
Seasonal Naive	~32 EUR/MWh	baseline
Ridge Regression	37.91 EUR/MWh	-15.5%
XGBoost + Two-Stage	**25.37 EUR/MWh (RMSE: 40.0)**	**+22.7%**

Ridge underperforms naive because L2 penalty shrinks price_lag_168h toward zero — it serves as a linear interpretability benchmark only.

MAE by year (walk-forward 2022–2024):

Year	MAE	Context
2022	80.18	Ukraine crisis — prices €200–400/MWh
2023	52.54	Transition year
2024	**25.37**	Primary benchmark year

Directional accuracy: **82.9%**. Two-stage architecture: binary classifier routes 17.2% of hours to an extreme-event regressor trained with price-proportional sample weights; a dedicated negative-price classifier (scale_pos_weight = 18.2, summer recall 91.1%, full-year 88.0%) feeds probability as a feature into the main model.

Validation: 52-window expanding walk-forward; min train 365 days; step 7 days; test full 2024 (8,784 hours); zero shuffling, zero future leakage.

Conformal Prediction (O'Connor et al., 2025): Regime-conditioned split conformal prediction with distribution-free coverage — no Gaussian assumptions. Empirical 90% coverage: **91.0%**.

Full 2024 out-of-sample predictions in submission.csv (8,784 rows): point forecast, 7 quantiles, conformal intervals, regime label, and Dunkelflaute risk score per hour.

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3. Prompt Curve Translation

Horizon	Definition	Instrument
Prompt Day Baseload	Mean 24h	DE DA base block
Prompt Day Peak	Mean hours 08–20 Mon–Fri	DE DA peak block
Prompt Week Base	7-day mean	EEX Week Base Future
Prompt Month Base	Calendar month mean	EEX Month Base Future

All with 80% and 95% conformal prediction intervals.

Signal: LONG if fair-value estimate (FV) $>$ reference $+$ 0.5 $\times$ CP80_width; SHORT if FV $<$ reference $-$ 0.5 $\times$ CP80_width; NEUTRAL otherwise. HIGH conviction if CP80 width $<$ 80% of 7-day mean.

Regime 3 (Dunkelflaute, residual load above 55GW and wind below 5GW): LONG prompt peak. Nov 2024 event drove DA prices to €145/MWh — gas peakers earned a disproportionate share of annual wholesale revenue over 111 hours. Regime 0 (renewable glut, penetration above 80% on summer weekends): SHORT prompt baseload, long battery charge/discharge spread.

Invalidation: wind revision $>$20%; TTF move $>$€3/MWh overnight; French nuclear drop $>$2,000 MW; DA clearing $>2\sigma$ from forecast; residual load change $>$15%; unplanned outage $>$1,000 MW; regime flip pre-delivery; conformal band expands $>$50% vs 7-day mean.

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4. AI-Accelerated Workflow

Three Gemini 2.0 Flash integrations — all programmatic, logged to JSONL, auditable.

Component 1 — LLM Data QA: Gemini receives schema plus 10 sample rows plus statistics and proposes 20 electricity-market-specific validation rules executed programmatically as Python boolean conditions. Component 2 — Market Commentary: 150–180 word Bloomberg-style note generated from 24 pipeline metrics only, with anti-hallucination guard cross-checking every number at $\pm$0.5 tolerance. Component 3 — Ingestion Config: field documentation converted to structured JSON ingestion config, eliminating manual field mapping hardcoding. All prompts, responses, and outputs logged to JSONL under outputs/logs/.

Component	Logged	Hallucination Check	On Failure
QA Rules	Yes — JSONL	N/A	Log + continue
Commentary	Yes — JSONL	Yes — $\pm$0.5 tolerance	Flag + log
Config Gen	Yes — JSONL	N/A	Fallback defaults

References: O'Connor et al. (2025) Energy and AI; Marcjasz et al. (2023) Energy Economics; Weron (2014) IJF; Wood Mackenzie (2025); Timera Energy (2025); FfE (2026).